MA1522 Linear Algebra for Computing

Lesson 2: Gaussian Elimination and Solving Linear Systems

Practice Quizzes

- You are expected to attempt the practice quiz after each lecture video.
- These quizzes do not count towards the final grade.
- You may attempt them as many times as you like.
- The due dates are only a guide to help you pace yourself — the quizzes remain accessible after they are past due.

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson2` (e.g. `A1234567Z_lesson2.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 2**.
- Deadline: today, **10:30 AM**.
- If you do not know how to upload your submission, come to ask me after class.

What Is an Eligible Submission?

Only an **eligible** submission earns the participation mark. On top of the instructions above, it must meet all of the following:

- It is complete: it includes an attempt at every one of Questions 1 to 3.
- It is handwritten or typed by you — photographs of slides taken during class are not acceptable.
- It is submitted before the deadline. No exception.

It is graded on effort and attempt, not on correctness.

Do Not Share Submissions

Sharing your work for another student to submit — or submitting work obtained from another student — constitutes academic misconduct. All students involved will be subject to disciplinary action: **at a minimum, 5% deducted from the final course grade**, and possibly more.

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Basic rules:

- Please (mainly) ask the questions related to the course.

Example 1

Each matrix below is the RREF of the augmented matrix of a linear system. How many of them are **inconsistent**? How many have **exactly one solution**? How many have **infinitely many solutions**?

(a) $\left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{array}\right)$ (b) $\left(\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right)$ (c) $\left(\begin{array}{ccc|c} 1 & -3 & 0 & 4 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right)$

(d) $\left(\begin{array}{cccc|c} 1 & 0 & 2 & 0 & 1 \\ 0 & 1 & -1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right)$ (e) $\left(\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right)$

(a) **one solution** — every variable column is pivot. (b) **inconsistent** — the rightmost column is pivot, so a row reads $0 = 1$. (c) **infinitely many** — column 2 is non-pivot: one parameter. (d) **infinitely many** — column 3 is non-pivot: one parameter. (e) **one solution** — both variable columns are pivot; the zero row says nothing.

So: **one** inconsistent, **two** with exactly one solution, **two** with infinitely many.

Section 1.4 Gaussian and Gauss–Jordan Elimination

Lesson 1 said what a REF *is*; these algorithms produce one, for any matrix A.

Gaussian elimination $\longrightarrow$ a REF:

1. Find the leftmost nonzero column, say column j.
2. If its top entry is 0, interchange the top row with a row whose entry in column j is nonzero.
3. Add multiples of the top row to the rows below it, clearing every entry below the leading entry.
4. Cover the top row and repeat (1)–(3) on what remains.

Gauss–Jordan elimination $\longrightarrow$ the RREF:

5. Scale each nonzero row so that its leading entry is 1.
6. From the bottom nonzero row upwards, add multiples of that row to the rows above it to clear the rest of its pivot column.

Recall: A has many REFs but exactly one RREF. Two students with different-looking REFs can both be right; two students with different RREFs cannot.

Question 1

For each of the following linear systems, find the **RREF** of its augmented matrix, and hence write down its general solution.

(a)
$$\begin{cases} x + 2y + 3z = 4 \\ 3x - 4y + z = 2 \\ x + y - 5z = 3 \end{cases}$$

(b)
$$\begin{cases} x - y + 2z = 3 \\ 2x - y - z = 0 \\ 4x - 3y + 3z = 6 \end{cases}$$

Question 1(a) — solution

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 3 & -4 & 1 & 2 \\ 1 & 1 & -5 & 3 \end{array}\right) \xrightarrow[R_3 - R_1]{R_2 - 3R_1} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 0 & -10 & -8 & -10 \\ 0 & -1 & -8 & -1 \end{array}\right) \xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 0 & -1 & -8 & -1 \\ 0 & -10 & -8 & -10 \end{array}\right)$$

$$\xrightarrow{R_3 - 10R_2} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 0 & -1 & -8 & -1 \\ 0 & 0 & 72 & 0 \end{array}\right) \xrightarrow[\frac{1}{72}R_3]{-R_2} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 0 & 1 & 8 & 1 \\ 0 & 0 & 1 & 0 \end{array}\right)$$

$$\xrightarrow[R_1 - 3R_3]{R_2 - 8R_3} \left(\begin{array}{ccc|c} 1 & 2 & 0 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{array}\right) \xrightarrow{R_1 - 2R_2} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{array}\right)$$

The rightmost column is non-pivot, so the system is **consistent**; all three variable columns are pivot, so there is **no parameter**. The RREF gives the general solution directly, with no back substitution:

$$x = 2, \qquad y = 1, \qquad z = 0.$$

Question 1(b) — solution

This time let the machine compute the RREF. In MATLAB:

```
A = [1 -1 2 3; 2 -1 -1 0; 4 -3 3 6];
rref(A)
```

$$\texttt{ans} = \left(\begin{array}{ccc|c} 1 & 0 & -3 & -3 \\ 0 & 1 & -5 & -6 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

(MATLAB prints no vertical bar; the last column is still the constants.)

The zero row says the third equation carried no new information. The rightmost column is non-pivot, so the system is consistent; columns $1, 2$ are pivot and column 3 is not, so there is **one** parameter. Put $z = t$:

$$x = 3t - 3, \qquad y = 5t - 6, \qquad z = t, \qquad t \in \mathbb{R}.$$

`rref` returns a matrix; deciding what it *means* is still your job — and it is the whole of Section 1.5.

Challenge — code it yourself

Both algorithms are just loops. In any language you like, write a program that reads an augmented matrix and prints a REF (Gaussian) and the RREF (Gauss–Jordan). Write the loop yourself — no built-in `rref`.

Test it on Question 1(a) and 1(b).

Question 2

The augmented matrix of a system in $x_1, \ldots, x_5$ is

$$R = \left(\begin{array}{ccccc|c} 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 2 \end{array}\right).$$

1. Write down the general solution using x_3 and x_5 as parameters.
2. Can you take x_1 and x_2 as the parameters instead?
3. How about the pair x_2 and x_5?

Question 2 — solution: part (1)

R is in row-echelon form — but not reduced, since column 4 is a pivot column with a nonzero entry above its leading entry. No matter: a REF is enough to count.

The pivot columns are $1, 2, 4$. The rightmost column is non-pivot, so the system is **consistent**; columns $3, 5$ are non-pivot, so there are **two** parameters and the solution is **not unique**.

Taking $x_3 = s$ and $x_5 = t$, the rows read $x_4 + x_5 = 2$, $x_2 + x_4 = 0$, $x_1 + x_3 + x_5 = 1$, and working upwards gives

$$x_1 = 1 - s - t, \quad x_2 = t - 2, \quad x_3 = s, \quad x_4 = 2 - t, \quad x_5 = t.$$

Question 2 — solution: parts (2) and (3)

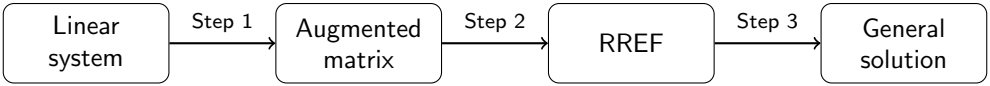
(2) Yes — the parameters need not be the non-pivot variables. Set $x_1 = s$ and $x_2 = t$. Then $x_4 = -x_2 = -t$, then $x_5 = 2 - x_4 = 2 + t$, then $x_3 = 1 - x_1 - x_5 = -1 - s - t$, so

$$x_1 = s, \quad x_2 = t, \quad x_3 = -1 - s - t, \quad x_4 = -t, \quad x_5 = 2 + t.$$

(3) No. The last two rows give $x_2 = -x_4$ and $x_4 = 2 - x_5$, hence $x_2 = x_5 - 2$. The two are locked together, so they cannot range independently — and indeed both answers above satisfy $x_2 = x_5 - 2$.

The same relations tie x_2, x_4, x_5 into a trio: fixing *any one* of them fixes the other two. So at most one of the three may be a parameter, and the forbidden pairs are exactly (x_2, x_4), (x_4, x_5), (x_2, x_5).

Summary — solving a linear system



- **Step 1.** Write down the augmented matrix.
- **Step 2.** Compute its RREF.
- **Step 3.** Read the solution off the RREF.

Remark

Computing the RREF and interpreting it is a basic skill we will need for almost every problem in the rest of the semester. Make sure you completely understand the whole process.

Question 3

Find all values of a and b for which the system in x, y, z

$$\begin{cases} ax + bz = 2 \\ ax + ay + 4z = 4 \\ ay + 2z = b \end{cases}$$

has (a) exactly one solution; (b) infinitely many solutions with one parameter; (c) infinitely many solutions with two parameters; (d) no solution.

Question 3 — solution: first two steps

$$A = \left(\begin{array}{ccc|c} a & 0 & b & 2 \\ a & a & 4 & 4 \\ 0 & a & 2 & b \end{array}\right) \xrightarrow{R_2 - R_1} \left(\begin{array}{ccc|c} a & 0 & b & 2 \\ 0 & a & 4-b & 2 \\ 0 & a & 2 & b \end{array}\right) \xrightarrow{R_3 - R_2} \left(\begin{array}{ccc|c} a & 0 & b & 2 \\ 0 & a & 4-b & 2 \\ 0 & 0 & b-2 & b-2 \end{array}\right) = B$$

Both operations are legal for *every* a — no division took place. But whether B is in row-echelon form depends on a, so the analysis splits in two.

Question 3 — solution: the two cases

Case $a \neq 0$**.** Then $B = \left(\begin{array}{ccc|c} a & 0 & b & 2 \\ 0 & a & 4-b & 2 \\ 0 & 0 & b-2 & b-2 \end{array}\right)$ is a REF. Columns $1, 2$ are pivot; column 3 is pivot exactly when $b \neq 2$; the last column is *never* pivot, so the system is always consistent.

Case $a = 0$**.** Columns 1 and 2 of B are zero, so the rows read $bz = 2$, $(4-b)z = 2$, $(b-2)z = b-2$. If $b = 2$ these become $2z = 2$, $2z = 2$, $0 = 0$: consistent, $z = 1$, with x and y free. If $b \neq 2$ the third row forces $z = 1$, and then the first row needs $b = 2$ — a contradiction.

$a \neq 0$, $b \neq 2$: exactly one solution;
$a \neq 0$, $b = 2$: infinitely many, one parameter;
$a = 0$, $b = 2$: infinitely many, two parameters;
$a = 0$, $b \neq 2$: no solution.

Challenge

Not one of these equations is linear, and yet:

$$\begin{cases} x^2 - y^2 + 2z^2 = 6 \\ 2x^2 + 2y^2 - 5z^2 = 3 \\ 2x^2 + 5y^2 + z^2 = 9 \end{cases}$$

Solve it.

Put $u = x^2$, $v = y^2$, $w = z^2$. The system becomes linear in u, v, w, and Gauss–Jordan gives $\left(\begin{array}{ccc|c} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right)$, so $u = 4$, $v = 0$, $w = 1$ and hence $x = \pm 2$, $y = 0$, $z = \pm 1$ — four solutions.

Four solutions, from a problem that is not a linear system: the "0, 1, or infinitely many" rule applies to the *linear* system in u, v, w, not to the original one.

Next lesson

- **Sections 2.1–2.3 Matrices and Matrix Operations**

Please watch the video clips for these sections beforehand.

If you do not know how to upload your submission, come to ask me after class.